Homework Week 11

Modern Physics

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For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

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SPECIAL RELATIVITY

- **Particle in track detector / Ex. 2.87**
  An Ω^- particle has rest energy $E_0 = 1672\,\text{MeV}$ and mean lifetime $\tau = 8.2 \times 10^{-11}\,\text{s}$. It is created and decays in a particle track detector and leaves a track $d = 24\,\text{mm}$ long. What is the total energy of the Ω^- particle?

- **Pair annihilation / Ex. 3.54**
  How much photon energy is required to produce a proton-antiproton pair? Where could such a high-energy photon come from?

- **Doppler effect: space transmission / Ex. 2.51**
  A spacecraft traveling out of the solar system at a speed of $v = 0.95\,c$ sends back information at a rate of $f_0 = 1400\,\text{kHz}$. At what rate f do we receive the information?

- **Relativistic energy / Ex. 2.68**
  The total energy of a body is found to be twice its rest energy $E = 2E_0$. How fast is it moving with respect to the observer?

- **Doppler effect / Ex. 2.92**
  A police radar gun operates at a frequency of $f = 10.5\,\text{GHz}$. The officer, sitting in a patrol car at rest by the highway, directs the radar beam toward a speeding car traveling $v = 80\,\text{mph}$ directly away from the patrol car. What is the frequency shift of the reflected beam, relative to the original radar beam?

NUCLEAR PHYSICS

- **Binding energy of the deuteron / Ex. 13.3 (Serway)**
  Calculate the binding energy of the deuteron (^2_1H), which consists of a proton and a neutron, given that the atomic mass of the deuteron is $M(^2_1\text{H}) = 2.014\,102\,u$, the mass of hydrogen, i.e., of the proton is $M(^1_1\text{H}) = m_\text{p} = 1.007\,825\,u$, the mass of the neutron is $m_\text{n} = 1.008\,665\,u$. The atomic mass unit (Dalton) is $u = 931.5\,\text{MeV}/c^2$.

- **Mass-energy imbalance / Ex. 2.84**
  The reaction $^2\text{H} + {}^2\text{H} \rightarrow n + {}^3\text{He}$ (where n is a neutron) is one of the reactions useful for producing energy through nuclear fusion.

  (a) Assume the deuterium nuclei (^2H) are at rest and use the atomic mass units of the previous problem and $M(^3\text{He}) = 3.016\,029\,u$.
  Note: You can use atomic masses for this calculation because the electron masses cancel out. This amount of energy is given up when this nuclear reaction occurs.

  (b) What percentage of the initial rest energy is given up?

EXTRA PROBLEMS

- **Relativistic momentum derivation / Ex. 2.73**
  Prove that $\beta = pc/E$. This is a useful relation to find the velocity of a highly energetic particle.

- **Relativistic momentum / Ex. 2.58**
  Find the force necessary to give a proton an acceleration of $a = 1 \times 10^{19}\,\text{m/s}^2$ when the proton has a velocity (along the same direction as the force) of (a) $v = 0.01\,c$, (b) $v = 0.1\,c$, (c) $v = 0.9\,c$, (d) $v = 0.99\,c$.

- **Relativistic Newton's second law**
  Consider Newton's second law, given by $\vec{F} = d\vec{p}/dt$ using the relativistic momentum $\vec{p} = \gamma m\vec{v}$.

  (a) Prove that $\vec{F} \parallel \vec{a}$ implies that either $\vec{F} \parallel \vec{v}$ or $\vec{F} \perp \vec{v}$ (Hint: expand the derivative on the right hand side, and rearrange the equation such that terms proportional to $\vec{F}$ or $\vec{a}$ are on the same side, and terms proportional to $\vec{v}$ on the other).

  (b) Express the acceleration $\vec{a}$ with $\vec{F}$ and $\vec{v}$ for general direction velocity and force. (Hint: Look for the acceleration in the form $\vec{a} = g\vec{F} + h\vec{v}$ where g and h are unknown scalar functions of $\vec{F}$ and $\vec{v}$. Substitute this back into the expanded Newton's law, and arrange all terms proportional to $\vec{F}$ and $\vec{v}$ to different sides. If $\vec{F}$

and $\vec{v}$ are not parallel, this equation can only be true if the coefficients of $\vec{F}$ and $\vec{v}$ are both zero, from this you can determine g and h. You should find a similar expression to the expanded Newton's law, but with the roles of $\vec{F}$ and $\vec{a}$ switched, and the m and γ factors arranged differently. Check that you recover the results for the special cases when $\vec{F}$ and $\vec{v}$ are parallel or perpendicular.)

- **Complicated Doppler effect / Ex. 2.53**
  Three radio-equipped plumbing vans are broadcasting on the same frequency f_0. Van 1 is moving North of van 2 with speed v, van 2 is fixed, and van 3 is moving West of van 2 with speed v. What frequency does van 3 hear from van 2? And from van 1?

- **Kaon decay / Ex. 2.96**
  A high-speed K^0 meson with rest energy $E_{0\,K^0} = 498\,\text{MeV}$ is traveling at a speed of $v = 0.90\,c$ when it decays into a π^+ and a π^- meson, both with with rest-energy $E_{0\,\pi^\pm} = 140\,\text{MeV}$. What are the highest and lowest speeds that the π mesons may have? (Hint: solve the problem first in the rest frame of the kaon for the magnitude of the π mesons' velocity, and consider that the direction in which the π mesons leave is indeterminate.)

- **Extrasolar planets / Ex. 2.105**
  One way astrophysicists have identified "extrasolar" planets orbiting distant stars is by observing redshifts or blueshifts in the star's spectrum due to the fact that the star and planet each revolve around their common center of mass (see Scientific American, August 2010, p. 41.). Consider a star the size of our sun ($M_s = 1.99 \times 10^{30}\,\text{kg}$) with a planet the size of Jupiter ($M_p = 1.90 \times 10^{27}\,\text{kg}$) in a circular orbit of radius $R_p = 7.79 \times 10^{11}\,\text{m}$ and a period $T_p = 11.9\,\text{y}$. (a) Find the speed of the star v_s revolving around the system's center of mass. (b) Assume that Earth is in the planet's orbital plane, so that at one point in its orbit the star is moving directly toward Earth, and at the opposite point, it moves directly away from Earth. How much is a light ray with wavelength $\lambda = 550\,\text{nm}$ redshifted and blueshifted at those two extreme points?